

# Evaluating ML-KEM at (much) higher security levels

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“Conservative” post-quantum security could mean avoiding algebraic structure vulnerable to future attacks [BCD<sup>+</sup>16], or maintaining a wide security margin against improving cryptanalysis. FrodoKEM, standardized by ISO [ISO26], provides the first by design, but widening its margin requires scaling parameters that already cost nearly ten times the bandwidth of NIST-standardized ML-KEM at the same security level [Nat24, GLN<sup>+</sup>25]. We show a module scheme can buy this margin cheaply: KAIBURR reaches a large security margin within FrodoKEM’s bandwidth while running roughly 3× faster. Starting from ML-KEM-1024 and fixing every other parameter, we vary the rank  $k$  and measure the decryption-failure probability using the Kyber scripts [CRY]. The  $2^{-128}$  bound holds at  $k = 7$  but fails at  $k = 8$ , so raising  $k$  further cannot buy higher security without sacrificing correctness.

Instead of changing the ring dimension  $n$  or modulus  $q$ , we propose a parameterized noise distribution  $f(t)$  on  $\{-2, \dots, 2\}$  with  $\Pr[\pm 2] = 2^{-t}$ ,  $\Pr[\pm 1] = \frac{1}{4}$ ,  $\Pr[0] = \frac{1}{2} - 2^{-(t-1)}$ . Its standard deviation  $\sigma(t) = \sqrt{0.5 + 2^{3-t}}$  decreases from 1 to  $\sqrt{0.5}$  as  $t$  grows. A sample is drawn from uniform bits  $a_1, \dots, a_t$  as  $x = (2a_t - 1)(1 - a_1 + 2 \prod_{i=1}^{t-1} a_i)$ , which is branchless; we machine-check both CT and speculative constant-time (SCT) in Jasmin [ABB<sup>+</sup>17].

Larger  $k$  raises decryption error, so we jointly raise  $t$  (lower  $\sigma$ ) to compensate, while favoring byte-friendly values of  $t$ .  $t = 4$  recovers ML-KEM’s sampler CBD(2),  $t = 8$  allows one-byte-per-coefficient SIMD packing, and  $t = 6$  gives more security per dimension and fewer PRF bytes at the cost of less regular packing. For each  $t$  we take the largest  $k$  meeting the failure bound and a communication cost under FrodoKEM-640 (9616 B pk, 9752 B ct).

**Table 1:** KAIBURR parameter sets. Classical/quantum security (Core-SVP) and failure probability computed with the Kyber scripts [CRY]; other parameters as in ML-KEM, compression removed.<sup>1</sup>

| Scheme   | $k$ | noise  | failure      | class. | quant. | pk (B) | ct (B) |
|----------|-----|--------|--------------|--------|--------|--------|--------|
| kaiburr4 | 7   | $f(4)$ | $2^{-133.7}$ | 479    | 434    | 2,720  | 3,072  |
| kaiburr6 | 18  | $f(6)$ | $2^{-134.1}$ | 1281   | 1162   | 6,944  | 7,296  |
| kaiburr8 | 24  | $f(8)$ | $2^{-139.9}$ | 1717   | 1557   | 9,248  | 9,600  |

KAIBURR has reference and AVX2 implementations in both C and Jasmin, available [here](#). We benchmark on two Intel generations following the methodology of [ABB<sup>+</sup>23]: single-core with Hyper-Threading and Turbo Boost disabled, medians over 10 001 runs with randomness from Linux `getrandom` (Table 2).

**Table 2:** KAIBURR (AVX2) vs. FrodoKEM-640-SHAKE (AVX2), CPU cycles; KAIBURR8 runs  $\sim 2.6$ – $3.1\times$  faster than Frodo. Jasmin stays within 17% of C, faster on Coffee Lake and slower on Raptor Lake.

| Microarch.               | Scheme                          | Impl.  | keypair   | enc       | dec       |           |
|--------------------------|---------------------------------|--------|-----------|-----------|-----------|-----------|
| Coffee L.<br>(i7-8700)   | kaiburr4                        | C      | 155,113   | 159,177   | 171,924   |           |
|                          |                                 | Jasmin | 144,489   | 145,158   | 150,135   |           |
|                          | kaiburr6                        | C      | 806,319   | 815,330   | 842,327   |           |
|                          |                                 | Jasmin | 762,472   | 770,586   | 781,276   |           |
|                          | kaiburr8                        | C      | 1,368,794 | 1,378,658 | 1,416,170 |           |
|                          |                                 | Jasmin | 1,294,067 | 1,303,213 | 1,311,650 |           |
|                          | FrodoKEM-640-SHAKE <sup>2</sup> |        | [GLN+25]  | 3,898,666 | 4,063,179 | 3,939,567 |
|                          |                                 |        |           |           |           |           |
| Raptor L.<br>(i9-13900K) | kaiburr4                        | C      | 120,886   | 124,488   | 134,556   |           |
|                          |                                 | Jasmin | 141,676   | 143,442   | 148,714   |           |
|                          | kaiburr6                        | C      | 647,542   | 651,956   | 676,062   |           |
|                          |                                 | Jasmin | 746,326   | 750,672   | 763,608   |           |
|                          | kaiburr8                        | C      | 1,095,258 | 1,103,782 | 1,133,918 |           |
|                          |                                 | Jasmin | 1,265,650 | 1,271,618 | 1,285,842 |           |
|                          | FrodoKEM-640-SHAKE              |        | [GLN+25]  | 3,121,248 | 3,246,262 | 3,189,934 |

We also implement and report KAIBURR on Pavona’s Asymmetric Cryptography Coprocessor (ACC) [Glo26], which reuses ML-KEM’s arithmetic core. Our simulation model raises DMEM from 32 to 128 KiB for large  $k$ .

**Table 3:** Per-operation cost of KAIBURR on ACC, in clock cycles; single deterministic cycle-accurate simulator run.

| Scheme   | keypair   | enc       | dec       |
|----------|-----------|-----------|-----------|
| kaiburr4 | 240,816   | 239,348   | 257,474   |
| kaiburr6 | 1,453,490 | 1,424,342 | 1,467,109 |
| kaiburr8 | 2,505,793 | 2,449,864 | 2,506,070 |

KAIBURR shows that a structured KEM can reach conservative security margins cheaply and within FrodoKEM’s bandwidth, across software and dedicated PQC hardware.

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<sup>1</sup>Removing compression allows us to obtain formally verified correctness bounds as in [BKHL<sup>+</sup>25]; this is left as future work.

<sup>2</sup>We modified the Frodo harness (<https://github.com/microsoft/PQCrypto-LWEKE>) to take the median of 10 001 runs for consistency.

## References

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